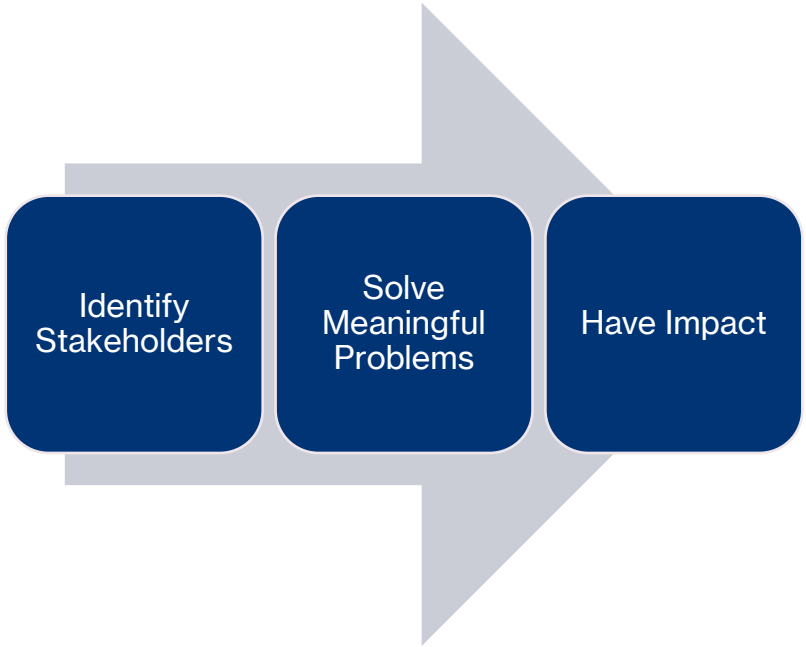
VALUE CREATION

Creating Impact Through Engineering Design

VALUE CREATION

- Why should anyone care about your project?
- Engineering is about creating value for people, not simply building technology
- Successful projects identify stakeholders, solve meaningful problems, and create measurable impact



COURSE PROJECT PHILOSOPHY

- Technical excellence is important but not sufficient
- Projects should address a genuine need and create value for one or more stakeholders
- Value creation should guide project selection, design decisions, and final presentations

WHAT IS VALUE?

Value is the benefit created for a stakeholder as a result of an action, decision, or solution.

- Value is not limited to money or profit
- Engineers create value by improving outcomes, reducing risk, saving resources, or enabling access

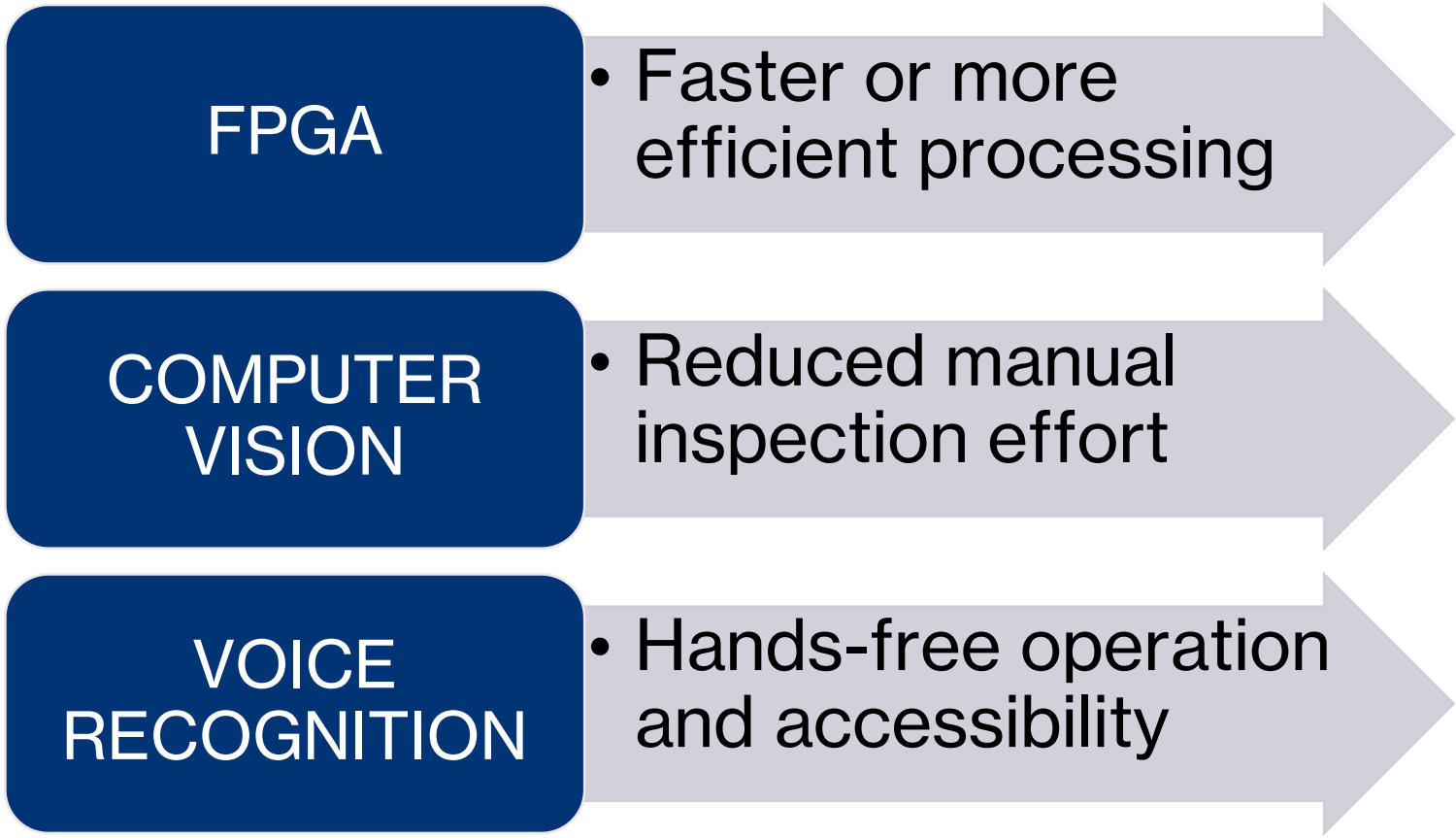
A SIMPLE TEST FOR VALUE

- Something is better for someone because of what you did
- If no stakeholder benefits, value may not have been created
- Always ask: who benefits and how?

ENGINEERING IS MORE THAN CREATING TECHNOLOGY

- Technology explains HOW a solutions works
- Value explains WHY the solution matters
- A project description should include **both** technology and value

EXAMPLES


FPGA
• Faster or more efficient processing
COMPUTER VISION
• Reduced manual inspection effort
VOICE RECOGNITION
• Hands-free operation and accessibility

WHO ARE YOUR STAKEHOLDERS?

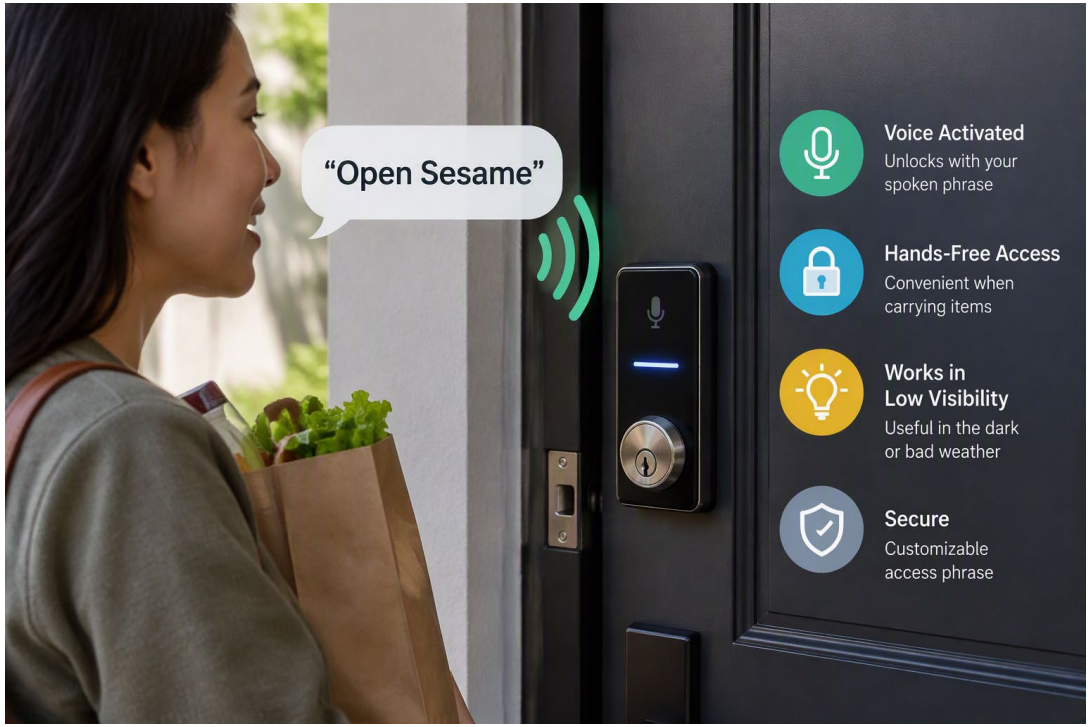
Examples of potential stakeholders:

- End users
- Customers
- Business owners/leaders
- Operators and maintenance personnel
- Researchers and educators
- Society and public agencies

STAKEHOLDER EXAMPLE

Phrase-Activated Lock

- Potential stakeholders: homeowners, apartment residents, accessibility users.
- Value: hands-free access when carrying items or in low visibility conditions.



TYPES OF VALUE

TECHNICAL

- Performance, accuracy, reliability

ECONOMIC

- Cost, Labor, Energy Savings

RISK REDUCTION

- Safety and Reliability Improvements

KNOWLEDGE

- Learnning and Understanding

TEAM/PROCESS

- Workflow and Communication

USER VALUE

- Convenience and Usability

MULTIPLE TYPES OF VALUE IN ONE PROJECT

Example: FPGA Guitar Pedal Board

- **Economic value:** lower cost than multiple pedals.
- **User value:** single integrated device (portability/convenience)
- **Technical value:** FPGA implementation (faster processing)



STRONG VS WEAK VALUE STATEMENTS

Strong statements identify stakeholders **and** benefits

Weak: “Our project uses computer vision”

Weak: “Our project uses machine learning”

Strong: “Our project helps homeowners improve security with less effort”

Strong: “Our project helps individuals enter secured spaces hands-free when carrying items, improving convenience and accessibility”

IMPROVING A VALUE PROPOSITION

- Who benefits?
- What changes for them?
- Why is that improvement meaningful?
- Can the value be measured or observed?

VALUE STATEMENT TEMPLATE

Our (project) helps _____by _____so they can _____

Every proposal should be able to complete this sentence clearly.

PROJECT PROPOSAL REALITY CHECK

- Have you identified a stakeholder?
- Is there a real problem?
- Can you explain the value?
- Would the stakeholder care?
- Can the project realistically deliver that value?

KEY TAKEAWAYS

- Start with people and needs
- Identify stakeholders early
- Consider multiple forms of value
- Communicate impact, not just features
- Build something that creates value